Talcher Coalfield

🏭 Introduction

The **Talcher Coalfield** is one of the **largest and most important coalfields in India**, located in Odisha. It is primarily operated by Mahanadi Coalfields Limited, a subsidiary of Coal India Limited.

Talcher is known for its **huge reserves of non-coking (thermal) coal**, making it a major contributor to India's electricity generation. The coalfield hosts several large opencast mines such as Jagannath, Lingaraj, and Kalinga, and plays a critical role in supplying coal to thermal power plants across the country. Due to its vast reserves and high production, Talcher is considered a **key energy hub of India**.

🌍 Geographical Overview – Talcher Coalfield

The Talcher Coalfield is located in the **Angul district** of Odisha and forms part of the **Mahanadi basin coalfields**, one of India's richest coal-bearing regions.

📍 Location & Setting

- Situated in the **Talcher region of Angul district**
- Lies in eastern India within a major industrial belt
- Well connected by rail and road networks for coal transportation

🌄 Physical & Geological Features

- Part of the **Gondwana coalfields**, containing extensive coal deposits
- Features **thick and gently dipping coal seams**, ideal for opencast mining
- Coal type: **non-coking thermal coal**

🌊 Nearby Natural Features

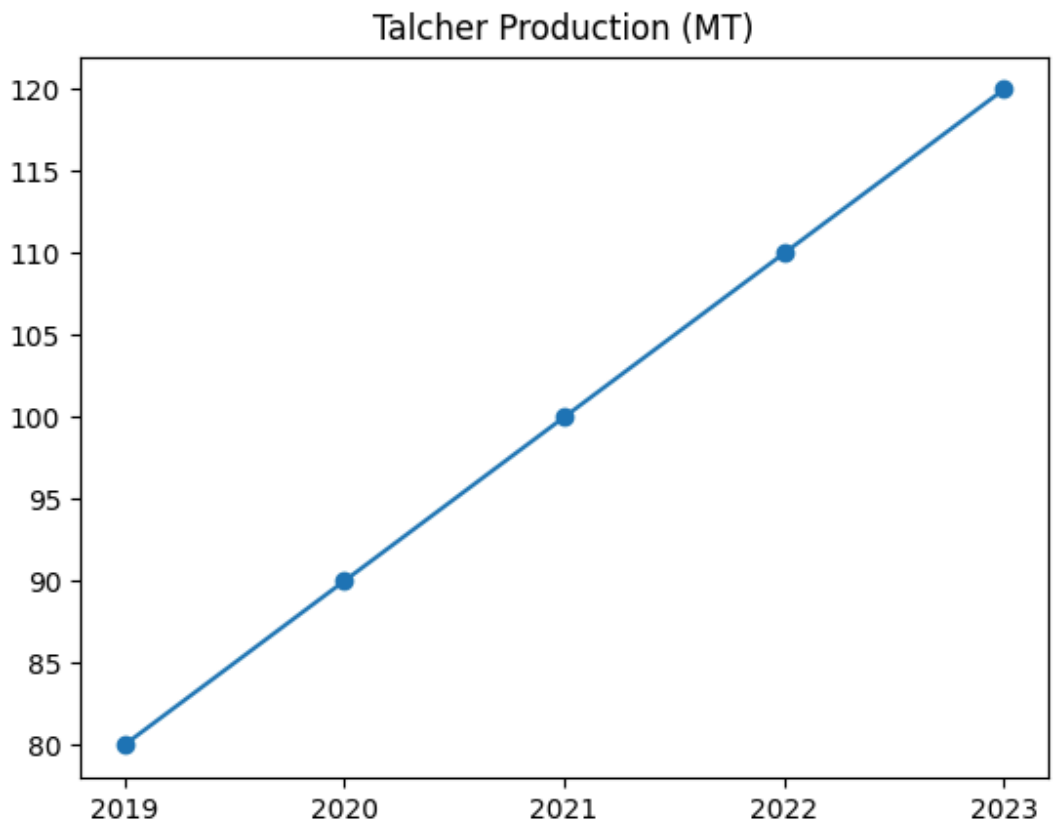
- Located near the Brahmani River basin
- Surrounded by major thermal power plants and industrial areas

🌦️ Climate

- Tropical climate
- Hot summers (up to ~45°C)
- Moderate to heavy monsoon rainfall

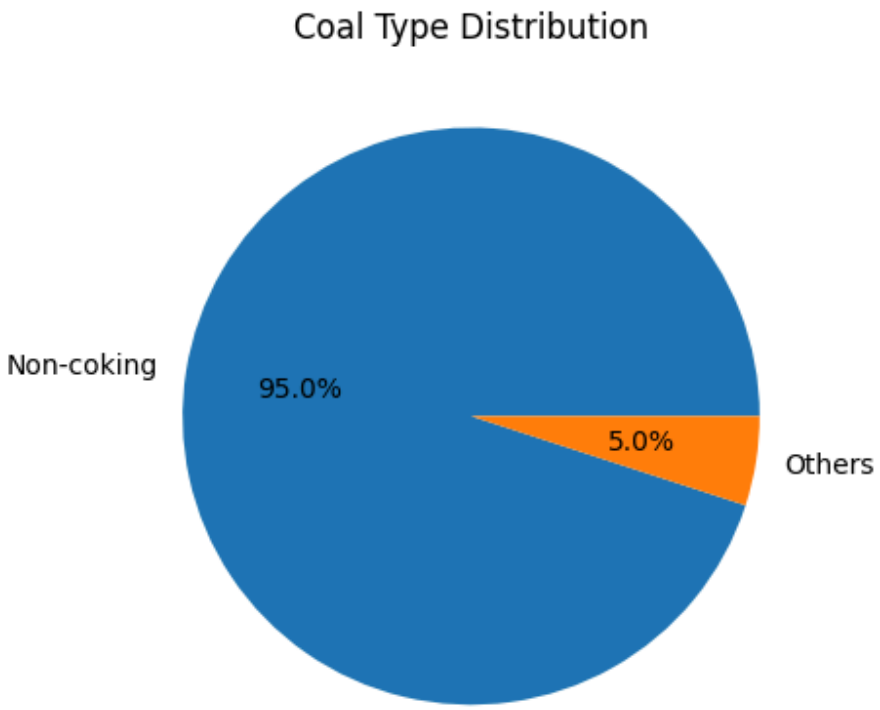
- Mild winters

📊 1. Production Trend (MT)



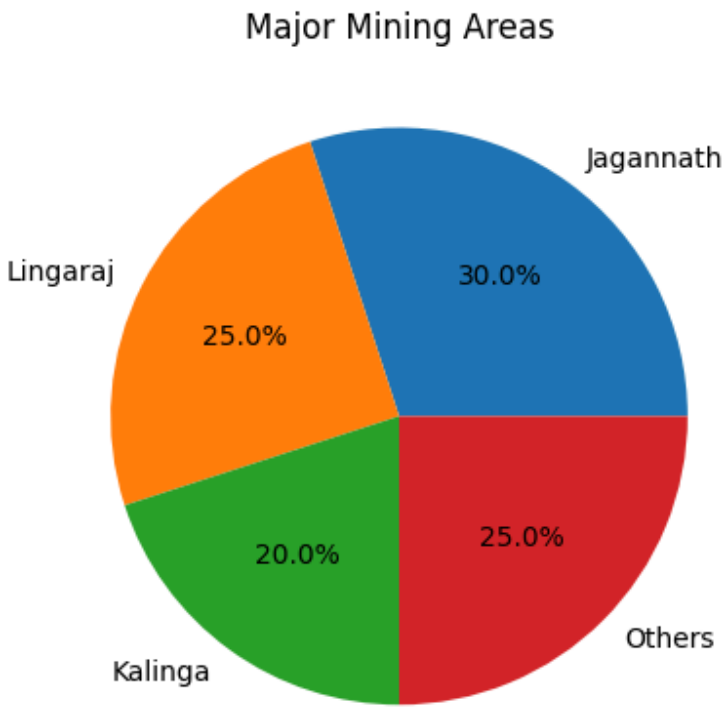
This chart shows the steady increase in coal production in Talcher coalfield.
The growth reflects expansion of large opencast mines and rising energy demand.
Significance: Useful for forecasting production and understanding Talcher's contribution to national coal output.

📊 2. Coal Type Distribution



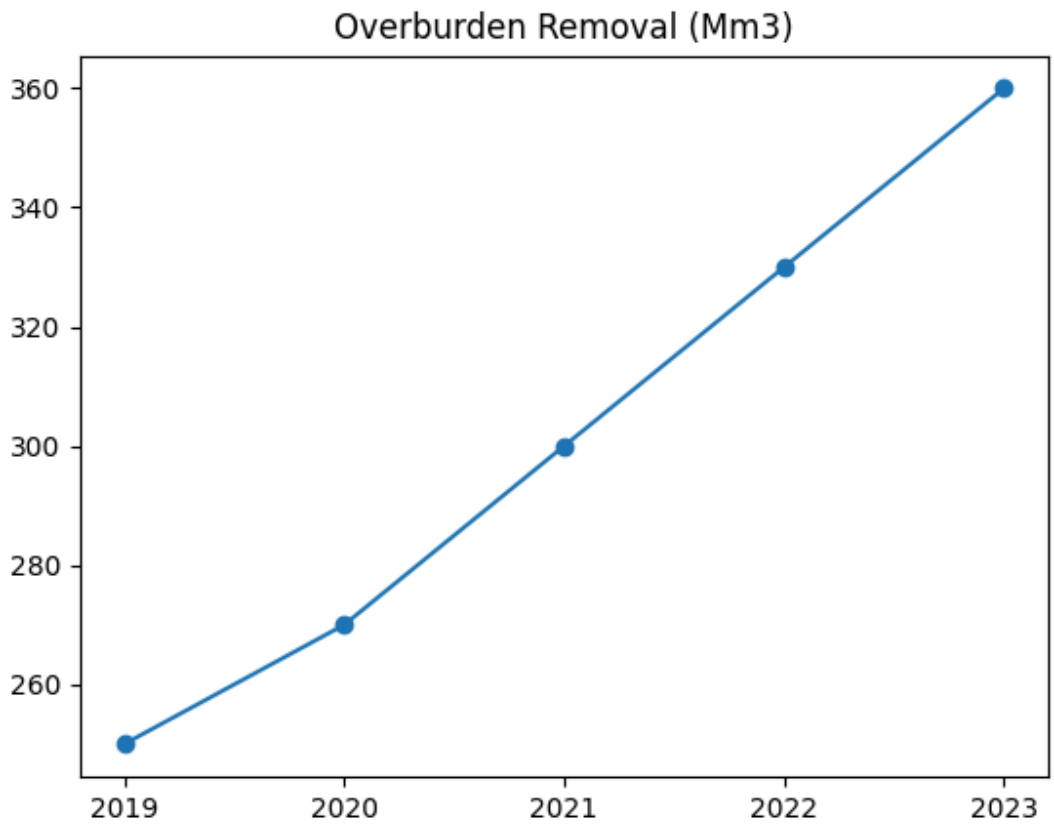
Shows dominance of non-coking (thermal) coal.
This type is mainly used in power generation.
Significance: Determines end-use sector and highlights importance in electricity production.

📊 3. Major Mining Areas



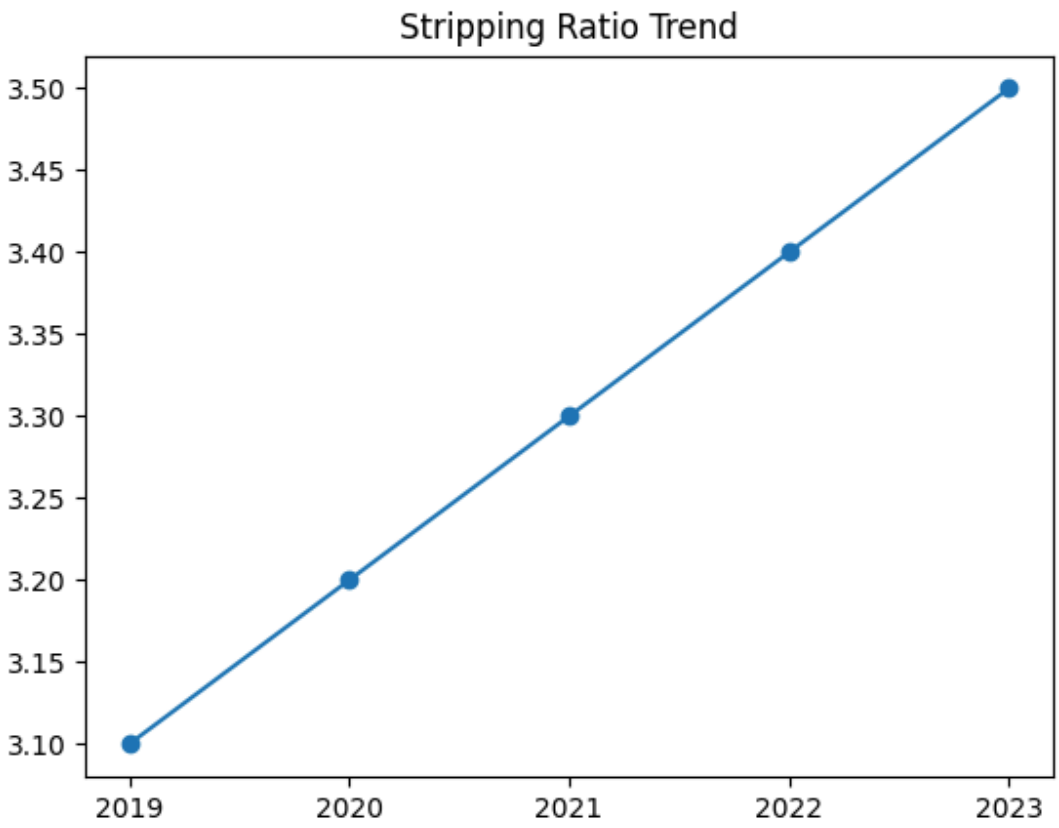
Represents contribution of major mines like Jagannath, Lingaraj, and Kalinga.
Different mines contribute based on their capacity and reserves.
Significance: Helps identify key production zones within the coalfield.

📊 4. Overburden Removal (Mm³)



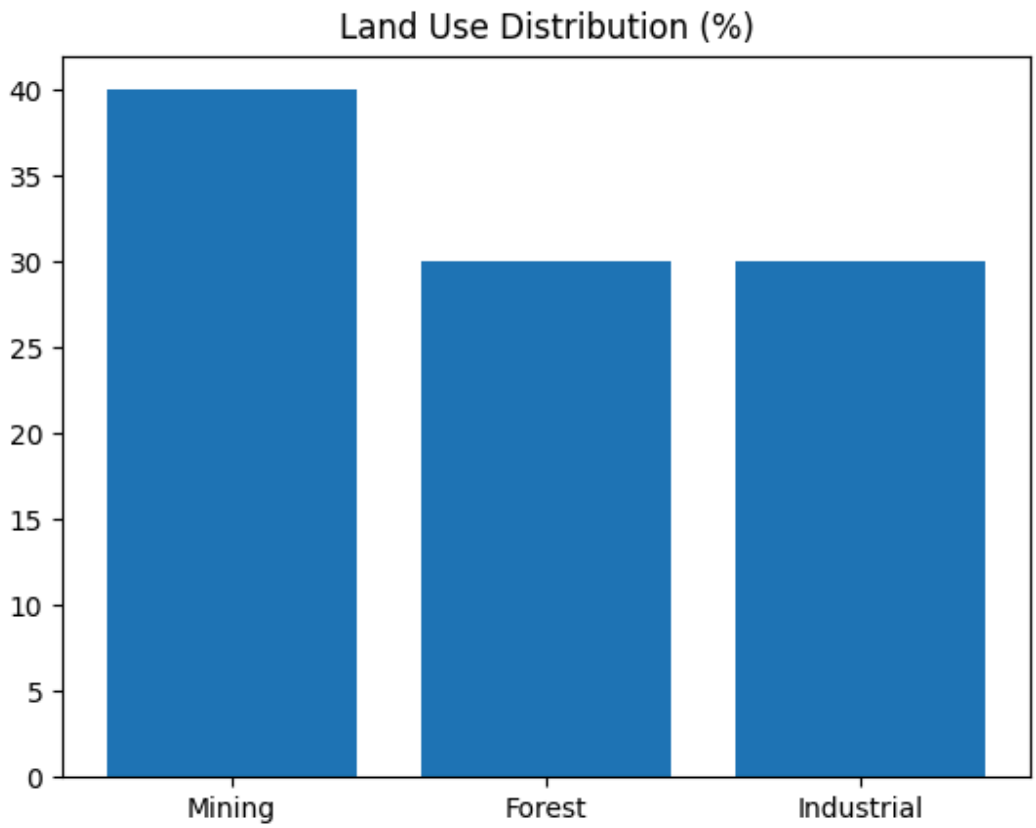
Displays increasing volume of overburden removal over time.
Indicates expansion and deeper mining operations.
Significance: Important for dump design, equipment planning, and cost estimation.

📊 5. Stripping Ratio Trend



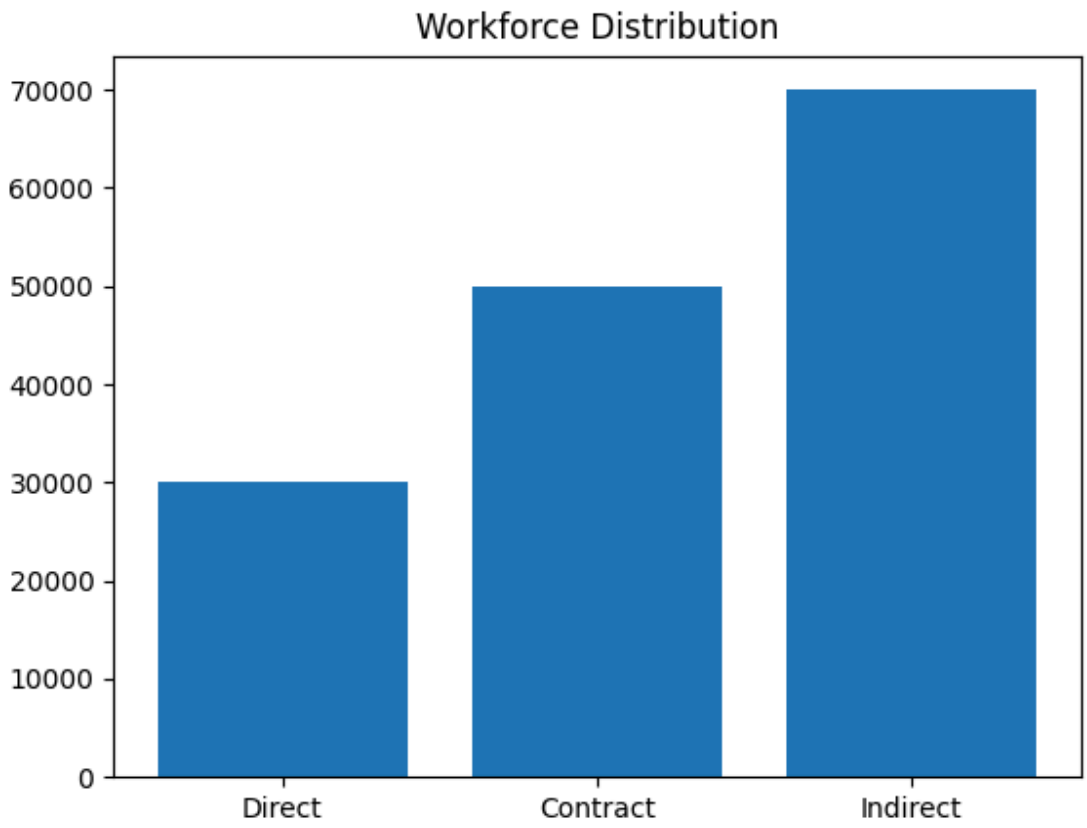
Shows gradual increase in stripping ratio.
Reflects deeper mining and higher operational cost.
Significance: Key parameter for economic feasibility and mine planning.

📊 6. Land Use Distribution



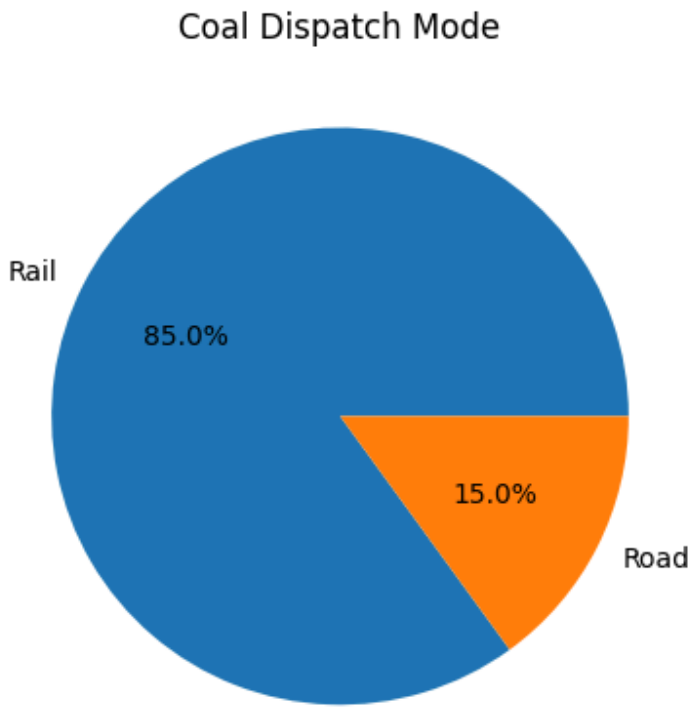
Represents distribution of land among mining, forest, and industrial use.
Talcher is a heavily industrialized coal region.
Significance: Important for environmental management and land reclamation.

📊 7. Workforce Distribution



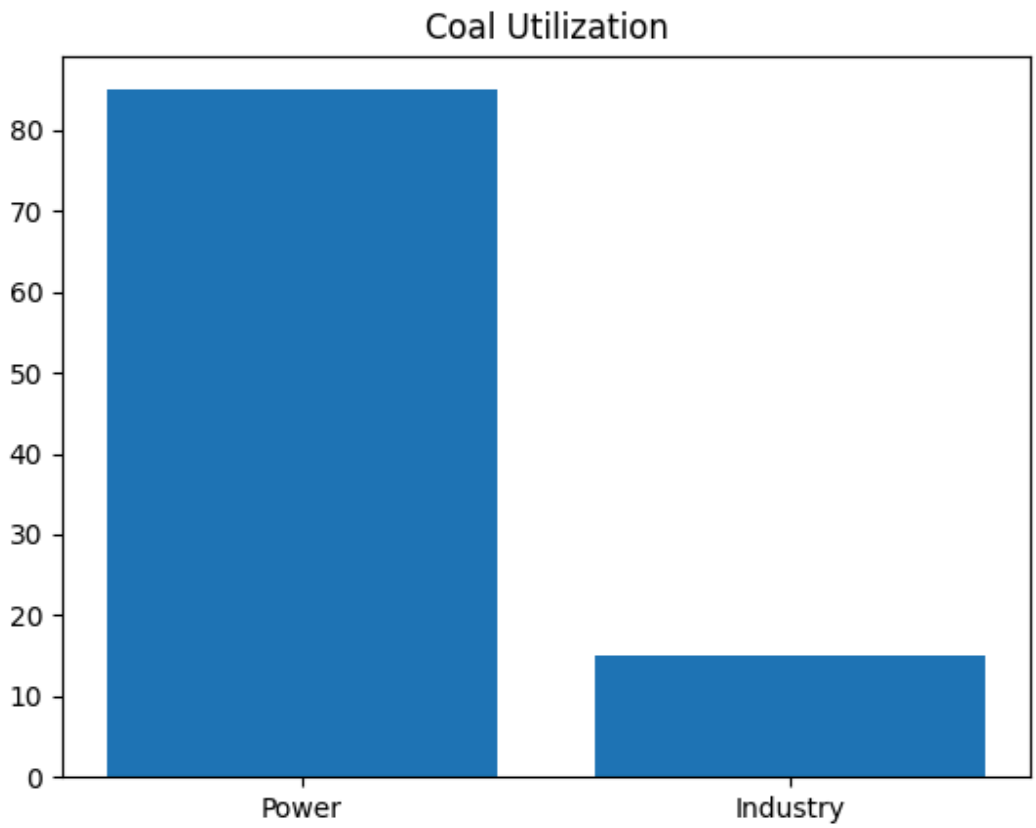
Displays manpower distribution across different employment categories.
Mining supports large-scale employment in the region.
Significance: Useful for manpower planning and socio-economic analysis.

📊 8. Coal Dispatch Mode



Represents transportation of coal mainly via rail.
Rail dominates due to bulk handling efficiency.
Significance: Important for logistics planning and cost reduction.

📊 9. Coal Utilization



Shows that most coal is used in thermal power generation.
Power sector consumes the majority of output.
Significance: Highlights Talcher's role in India's energy supply.